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Experimental investigation of specific heat and thermal stability of MOF-303 at different water vapor loading and various temperatures

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ABSTRACT

Measuring the specific heat capacity (Cp) of materials is essential for optimizing thermal management in various applications. Cp directly influences the heat storage and heat transfer performance of materials. Despite the widespread study of Metal Organic Frameworks (MOFs), the Cp of MOF-303, known for its high water adsorption capacity; has not been previously measured. This study focuses on the measurements of Cp for MOF-303 across a temperature range of 10 °C to 95 °C and at various water vapor loadings. Using Differential Scanning Calorimetry (DSC) and Thermogravimetric Analysis (TGA), we found that the dry MOF-303 exhibited a Cp of approximately 0.9175 J/g.°C at 10 °C, increasing to around 1.1695 J/g.°C at 95 °C. In contrast, fully saturated MOF-303 samples showed higher values, with Cp ranging from 1.72 J/g.°C at 10 °C to 2.41 J/g.°C at 95 °C. These findings demonstrate that water saturation significantly enhances Cp, making MOF-303 highly efficient for adsorption cooling systems. Thermal cyclic stability tests confirm the robustness of MOF-303 under repeated heating and cooling cycles. These results provide critical insights into the thermal behavior of MOF-303 and offer a novel contribution to the field. Cp correlations are supplied as a function of temperature and water vapor uptake, facilitating more accurate modeling and optimization of energy-efficient cooling applications.

1. Introduction

The specific heat at constant pressure (Cp) refers to the amount of heat necessary to raise the temperature of a unit mass of a substance by one degree Celsius (or one Kelvin) while maintaining constant pressure [1]. As a fundamental thermodynamic property, Cp provides insights into the behavior of materials under various thermal conditions and is essential for calculating other thermodynamic quantities like enthalpy, entropy, and Gibbs free energy. Understanding Cp is also valuable in selecting materials that efficiently store or transfer heat. Additionally, measuring Cp can optimize energy use in processes that involve heating or cooling. For instance, in industrial applications, knowing the Cp of materials can help lower energy costs by minimizing the heat needed to achieve the desired temperature. Moreover, measuring Cp contributes to understanding fundamental properties [2], such as molecular motion and heat capacity, which can drive advances in fields like condensed matter physics, chemistry, materials scienceand other thermal desalination systems and other renewable systems [3–6].

Differential Scanning Calorimetry (DSC) [7] is the most commonly used method to measure Cp due to its high sensitivity and accuracy in detecting phase transitions, heat capacity changes, and thermal stability across a broad temperature range. DSC provides precise control over temperature conditions, making it ideal for studying the unique thermal properties of Metal Organic Frameworks (MOFs). Complementary techniques such as Temperature-Modulated DSC (TM-DSC) enhance resolution for complex thermal events [8], while methods like Adiabatic Calorimetry and Thermogravimetric Analysis (TGA) provide additional insights into heat capacity and stability, supporting DSC's findings and offering a more comprehensive thermal analysis of these porous materials. McHugh et al. [8] compared (TM-DSC) and standard DSC for

Abbreviations: ACS, Adsorption Cooling Systems; BET, Brunauer–Emmett–Teller (surface area measurements); CNT, Carbon Nanotube; Cp, Specific Heat at Constant Pressure; DVS, Dynamic Vapor Sorption; DSC, Differential Scanning Calorimetry; MOF, Metal Organic Frameworks; PSD, Pore Size Distribution; P-XRD, Powder-X Ray Diffraction; SEM, Scanning Electron Microscope; TGA, Thermogravimetric Analysis; TM-DSC, Temperature Modulated Differential Scanning Calorimetry.

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measuring Cp during the isothermal curing of an epoxy. TM-DSC has provided detailed data by separating reversible and non-reversible heat flows but is complex and time-intensive, while standard DSC has offered quicker, simpler estimates for fully cured samples. Results have shown that Cp values obtained from both methods correlate well, with variation in Cp during cure remaining constant around 17–24 % of the minimum value. Standard DSC can serve as a practical alternative to TM-DSC for preparing experiments and initial estimates. In terms of DSC, different schemes could be applied to measure Cp like the direct method, Saphire method, and Iso Step DSC, …etc.

The Direct and Saphire techniques are carried out using a conventional DSC device and a linear temperature scheme. With conventional DSC, one total heat flow is sent and received from the sample. However, the heat capacity contains two parts: reversing heat flow (Cp) and non-reversing heat flow (Latent heat). Melting, chemical reactions, crystallization, and other physical or chemical changes are related to the latent heat capacity. On a DSC curve, these thermal processes manifest as endothermic or exothermic peaks. As a result, for endothermic activities, the latent heat capacity is plus, while for exothermic acts, it is minus [7]. On the other hand, the quantity of heat adsorbed as a result of molecular rearrangements and general movement is known as the sensible heat capacity. This element is positive. Assuming no thermal events take place, the diagram's DSC curve demonstrates a straight relationship between the sensible heat capacity and the measured heat flow. The sensible heat capacity establishes the base of other associated peaks in numerous transitions [7].

Regarding, adsorption cooling cycles, water harvesting devices, and other thermal applications involving porous materials, specifically solid desiccant materials like (MOFs) adsorbents, the specific heat capacity is one of the two main thermal properties influencing the beds [9], the other being adsorption isosteric heat [10–13]. Understanding the specific heat capacities of MOFs across the operating temperature range is vital for accurately forecasting the thermal behavior of many applications like adsorption cooling cycles. For example, the specific heat capacity (Cp) is a necessary parameter for calculating the energy balance of gases passing through a sorbent bed [14].

MOF-303, Metal Organic Framework composed of aluminum and 1,2,4-triazole, is recognized for its high water adsorption capacity [15], making it ideal for adsorption cooling cycles [16]. In these cycles, the cooling efficiency largely depends on the heat capacities of both the adsorbent like MOF-303, and the adsorbate (water) [17]. Knowing the specific heat (Cp) of MOF-303 at various stages of water loading allows engineers to design and optimize the cooling cycle for maximum energy efficiency. A higher Cp indicates that more energy is needed to change the temperature of the MOF, influencing the speed and effectiveness of the cooling process [14]. In studies of thermodynamic systems like adsorption cooling cycles, temperature-dependent models of specific heat have shown substantial improvements in accuracy over constant models, particularly in applications involving fluctuating temperature ranges and non-ideal conditions [18]. Al-Sarkhi et al. [18] demonstrated that higher-order polynomial models significantly enhance the prediction accuracy for specific heat in engine cycles. This insight underlines the need for precise specific heat models when examining materials like MOF-303, whose thermal performance may vary with temperature. Optimization of thermal properties, such as specific heat capacity, is critical in maximizing the energy efficiency of heat transfer systems. Okonkwo et al. [19] demonstrated how nanofluids with tailored thermal properties can enhance heat transfer performance in flat plate collectors. Their findings underscore the potential of MOF-303's high specific heat under varying moisture levels, which may be exploited to improve adsorption cooling or energy harvesting applications. Based on the experimental results, machine learning has emerged as an effective tool for predicting thermophysical properties. Jamei et al. [2] successfully applied models such as Kernel Extreme Learning Machines (KELM) to predict specific heat capacity in metal oxide nanofluids. Applying similar models to MOF-303's specific heat across different humidity levels could refine predictions and improve thermal management in adsorption cooling applications. Advanced machine learning techniques, such as ensemble learning, have shown high accuracy in predicting the specific heat capacity of nanofluids, as demonstrated by Deymi et al. [20]. Implementing similar models could aid in developing accurate predictive models for MOF-303, enhancing its application in temperature-sensitive systems. Thermal stability in materials with distinct phase-specific heats is crucial for their reliability in thermal cycling applications. Chakraborty [21] proposed an enthalpy-porosity model that captures the effect of phase-specific heat differences, a consideration that could enhance understanding of MOF-303's performance in applications with cyclic thermal loads.

While the literature is replete with MOFs' synthesis, gas adsorption isotherms, and isosteric heat data, measurements on their specific heats are far less common. Understanding the specific heat at different water loadings provides insight into how MOF-303's structure and performance might evolve over time or under varying operational conditions, which is crucial for ensuring long-term stability in adsorption cooling applications. The Cp of MOF-303 varies depending on the amount of water adsorbed, directly affecting its cooling capacity and efficiency. Measuring Cp across different water loadings enables a better understanding of how the material's heat capacity changes. Accurate Cp data is vital for developing reliable models and simulations of the adsorption cooling process, which help predict the behavior of MOF-303 under different environmental conditions, thus optimizing the design and control of the cooling system.

This study aimed to measure the Cp of MOF-303 across a broad range of temperatures and at varying levels of water vapor loading. Additionally, the study investigated the thermal cyclic stability of these Cp values using the Saphire method through cyclic linear programming of successive heating and cooling.

To achieve these objectives, the specific heat capacities of MOF-303 were measured over a temperature range of 10–95 °C and at different degrees of water vapor loading, from the completely dried state of MOF-303 to its fully saturated state, utilizing a conventional differential scanning calorimeter device (DSC-60 Plus). The study employed the Sapphire method, following European Standards (DIN 51007) for a carefully designed experimental protocol.

This research is expected to be broadly beneficial for modeling adsorbent systems and particularly useful for understanding the thermal behavior of adsorption cooling cycles.

2. MOF-303 synthesis

MOF-303 was synthesized in King Fahd for Petroleum and Minerals Labs by a solvothermal method [12]. 1.74 g of 1-H-3,5-pyrazoledicarboxylic acid monohydrate ($H_2PZDC{\cdot}H_2O$) and 0.6 g of NaOH were combined with water, and then $AlCl_3{\cdot}6H_2O$ was added, resulting in white precipitates. After 24 h of heating at 100 °C, the mixture was cooled and then repeatedly cleaned with eth-anol and deionized water.

After drying, the product was heated to 80 °C in a vacuum oven to activate it. In terms of Characterization steps. Powder-X Ray Diffraction (P-XRD) was performed to confirm crystallinity, porosity analysis using nitrogen adsorption-desorption isotherms (surface area: 1250 m^2/g), Scanning Electron Microscopy (SEM) to visualize the crystalline structure, Thermo-Gravimetric Analysis (TGA) to determine thermal stability up to 400 °C, and Dynamic Vapor Sorption (DVS) to evaluate water vapor adsorbed by MOF-303, all characterization results are compared with other published data to assure obtain high quality of MOF-303 [15]. The detailed steps for MOF-303 synthesis and characterization are mentioned in the **supplementary data file.**

3. Experimental concept of measuring specific heat capacity

To estimate the value of specific heat for MOF-303, conventional DSC was utilized since it uses small samples, which is easier than the

modified temperature DSC device, and gives relatively good results [8]. Moreover, MOF-303 was synthesized using the solvothermal method on a small-scale procedure [16]. The DSC heat flow signal obtained by the sample test was analyzed by comparing it to the DSC signal obtained from the reference material test of well-known specific heat (Sapphire Al_2O_3). Both data sets were corrected using a baseline adjustment using an empty crucible which was determined in a separate test.

4. DSC description

The DSC-60 Plus, from Shimadzu, is a heat flux differential measurement device that works similarly to differential thermal analysis (DTA). As shown in Fig. 1, when the heater is powered on, the temperature of the heating block (Tb) is monitored and controlled according to a preset program. The heat from the heating block travels through a thermal resistance (R) to each part, causing the temperature of the sample section (Ts) and the reference section (Tr) to change in response to the programmed temperature (Tb). The temperature of the sample section (Ts) and the temperature difference between the sample and reference sections ($\Delta T = Ts - Tr$) are measured at various points during this operation.

A reference material is often selected based on how thermally inert it is within the temperature range under investigation.

A Chromel-Alumel thermocouple is used by the DSC-60 Plus to measure ΔT, and a Chromel-Constantan differential thermocouple is used to detect ΔT.

5. Experimental procedure

The experimental work was implemented in three main stages. The first stage represented the most crucial part of the experimental investigation. In this stage, MOF-303 samples were prepared by activation, and then water vapor adsorption up to saturation. The second stage was implemented using a TGA device to reduce the water content in MOF-303 samples before being sealed. The last stage was to measure Cp using conventional DSC-60 Plus. The main aim of the study was to determine Cp values at different water loadings using the Saphire method and investigate the thermal cyclic stability of Cp values using the Saphire cyclic method.

5.1. Preparing MOF-303 samples

1. A furnace maintained at 30 °C was evacuated, and a glass vessel containing initially hot water was placed inside to generate water vapor.
2. The MOF-303 sample was introduced into the furnace for approximately 12 h, positioned adjacent to the water vessel.
3. Subsequently, the saturated MOF-303 sample was transferred to a glass vessel and sealed in preparation for thermogravimetric analysis (TGA).
4. After taking the crucible within the TGA device, seven samples, each containing 10.5 mg of saturated MOF-303, were placed onto the crucible.
5. Each crucible was heated at a designated temperature to facilitate the reduction of water content in the MOF-303.
6. The free water sample was identified at 150 °C, while the saturated sample corresponds to the one initially removed from the oven.
7. The remaining MOF-303 samples are heated to varying temperatures to obtain different levels of water vapor content.
8. The mass of each crucible was recorded using a highly sensitive TGA calorimeter (with a precision of up to five decimal places in

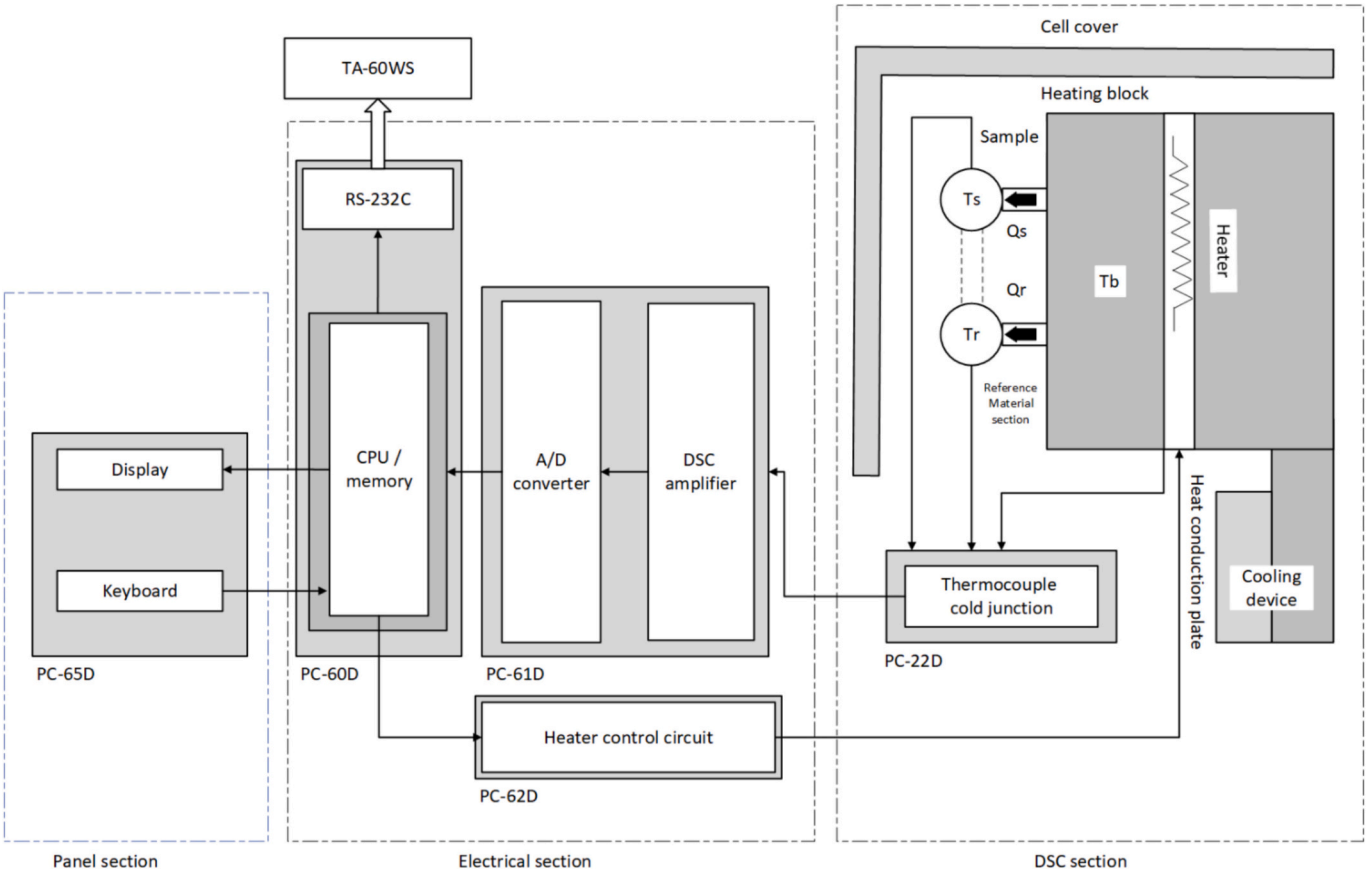


Fig. 1. DSC schematic diagram.

milligrams) and immediately sealed to preserve different levels of water vapor loading.
9. The loading ratios were calculated by referencing the mass of the first crucible removed from the furnace, which served as the weight reference for the saturated MOF-303, and the mass of the sample heated to 150 °C, which served as the reference for the pure dried MOF-303.

5.2. Differential Scanning Calorimetry (DSC)

Three steps were employed to measure the Cp of the prepared samples,

1. Place the MOF-303 sample at the given water vapor in the DSC pan, alongside an empty reference pan.
2. Run the DSC according to the specific temperature program as shown in Fig. 2. Two procedures are followed to calculate the specific heat by different methods like the Saphire method and the cyclic Saphire method as indicated in Fig. 2. The Saphire method is used to estimate the Cp values while the cyclic method is applied to measure the thermal stability of MOF-303.
3. Place a standard material of well-known specific heat (Sapphire Al_2O_3) along with the same empty and repeat the same procedures as shown in Fig. 2.
4. Place an empty crucible as a baseline to correct the results of the standard and samples along with the same empty and repeat the same procedures as shown in Fig. 2.

5.3. Thermo-gravimetric analysis (TGA)

A tared crucible containing 10.5 mg of saturated MOF-303 was heated from room temperature to 150 °C at a heating rate of 10 °C/min and then held at this temperature for 4 mins. During the process, Nitrogen gas was introduced at a flow rate of 10 L/min. The temperature of 150 °C was chosen as a reference point to represent a sample of MOF-303 that is free of water [12]. This initial stage aimed to determine the mass loss, allowing for the calculation of the water vapor released from the saturated sample. The mass loss observed up to 150 °C when compared with the initial mass indicated the amount of water vapor lost, from which the water uptake level was determined.

Subsequently, additional samples of MOF-303, each weighing 10.5 mg, were loaded into separate crucibles and heated to various temperatures (30, 40, 60, 80, and 100 °C) using the TGA. These varying temperatures were selected to regulate the amount of water vapor retained in the samples. Upon completion of the TGA analysis, each crucible was promptly sealed and stored in a plastic container before further examination with a differential scanning calorimetry (DSC) device. Fig. 3 shows the TGA curves obtained at different heating temperatures. The graph presents the results of a thermogravimetric analysis (TGA) for MOF-303, with each line representing a crucible containing 10.5 mg of the material. These crucibles were heated to various temperatures to alter the water vapor uptake before being sealed. The y-axis displays the mass, while the x-axis shows the temperature, ranging from 20 °C to 150 °C. Distinct colored lines represent the different temperature settings: 30 °C, 40 °C, 60 °C, 80 °C, 100 °C, and 150 °C. As the temperature increases, the mass of the material decreases, with a more rapid drop

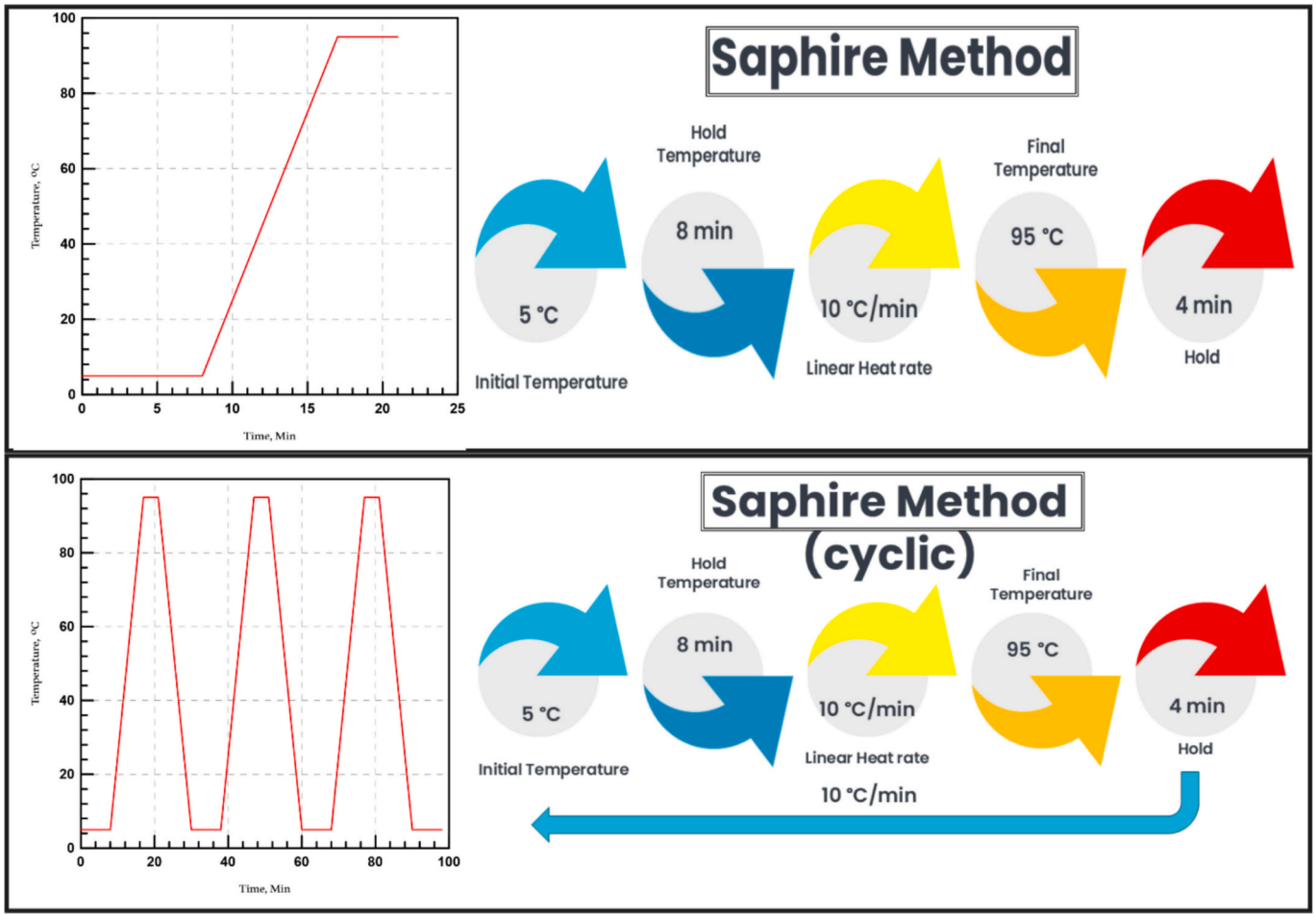


Fig. 2. Conventional DSC methods for measuring MOF-303's Cp.

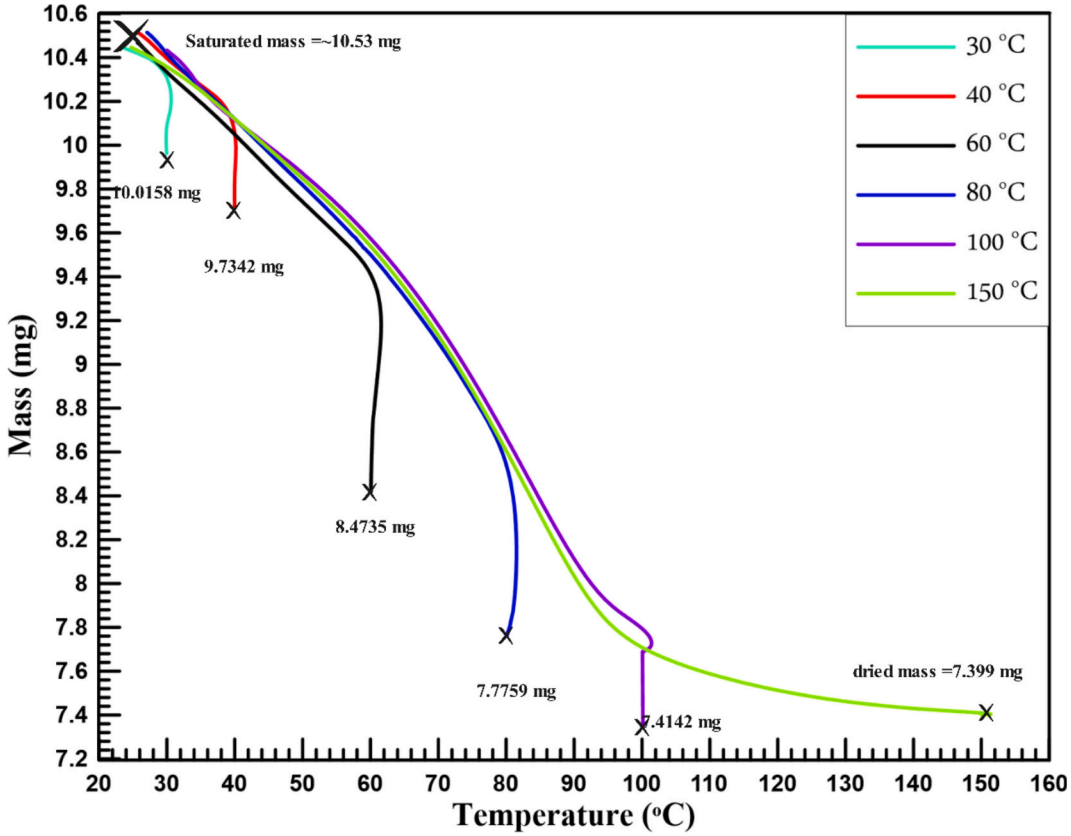


Fig. 3. TGA curve of MOF-303.

observed at higher temperatures. Notable points on the graph are annotated with specific mass values, starting with a saturated mass of 10.5 mg at 30 °C and reducing to a dried mass of 7.39 mg at 150 °C. it is worth mentioning that MOF-303 can adsorb up to 45 % of it is weight. Fig. 3 effectively illustrates how increasing temperatures influenced the water vapor content in the sealed crucibles containing MOF-303. To calculate the dimensionless uptake for each sample Eq. 1 was used and the degree of uptake was calculated based on Eq. 2. Table. 1 indicates the values for *q and dQ*.

$$q(T) = \frac{m(T) - m(150^{\circ}C)}{m(saturated) - m(150^{\circ}C)} \quad (1)$$

$$dQ = \frac{m(T) - m(150^{\circ}C)}{m(Saturated)} \quad (2)$$

Where are the equilibrium uptake and degree of uptake. q represents the dimensionless reference to describe the percentage of water vapor available for each sample after heating to temperature T concerning the total water vapor contained in the initial mass. dQ indicates a dimensionless reference for water vapor available at each sample heated to temperature T to initial MOF-303 mass. $m(150) = 7.399\,mg$, $m(saturated) = 10.5358\,mg$. $m(T)$ is the mass of MOF-303 plus water vapor.

6. Validation

To validate the differential scanning calorimetry (DSC-60 Plus) measurements, a standardized procedure was employed. Two crucibles, each containing 21 mg of alumina (Al_2O_3), were prepared. The DSC-60 Plus instrument was operated in two configurations: in the first, one alumina crucible was placed alongside an empty crucible; in the second, one crucible containing alumina served as the reference, while the other served as the sample. Following the measurements, the specific heat data obtained were compared with reference data for sapphire (Al_2O_3), Table 2 indicates the difference between the published and measured values for Al2O3 heat capacities, maximum deviation of 2.38 % was observed. Moreover, as illustrated in Fig. 4, the two graphs were nearly identical from 30 °C to 65 °C. This consistency in the latter temperature range confirms the accuracy and reliability of the DSC-60 Plus instrument, and the measurement procedure, for measuring the specific heat

Table 1
MOF-303 samples information.

TGA temperature before sealing the crucibles, °C	Mass, mg	q, Uptake	Water vapor content, mg/10.5 mg of MOF-303	dq, Degree of uptake, %
Saturated	10.5358	1.0000	3.14	29.82
30.00	10.0158	0.8342	2.62	24.87
40.00	9.7342	0.7444	2.34	22.20
60.00	8.4735	0.3425	1.07	10.21
80.00	7.7759	0.1202	0.38	3.58
100.00	7.4142	0.0048	0.02	0.14
150.00	7.3990	0.00	0.00	0.00

Table 2
Comparison between the measured and reference values of Cp for Al_2O_3.

Temperature, °C	Reference values, J/g. °C	Measurement values, J/g. °C	Error, (%)
30	767.649	757	2.115695
35	778.1355	766	1.963771
40	788.312	775	1.814919
45	798.1785	783	1.794469
50	807.735	792	1.650246
55	816.9815	800	2.287167
60	825.918	807	2.383847
65	834.5445	820	1.759847

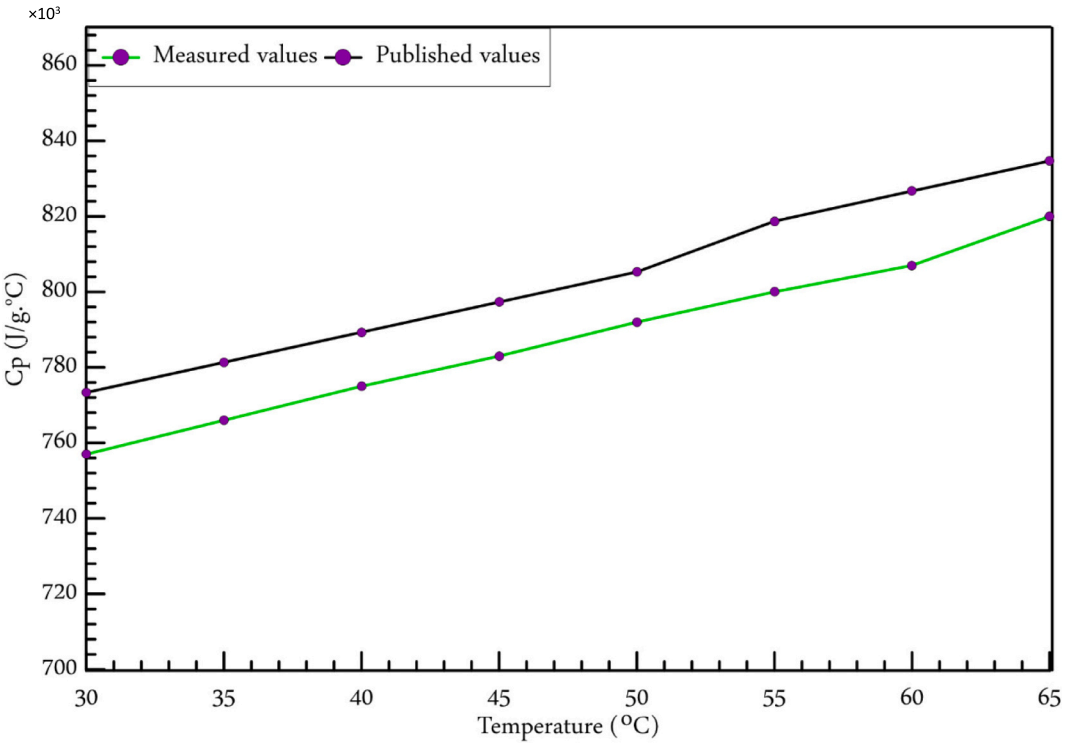


Fig. 4. DSC validation.

of Al_2O_3.

While traditional TGA-DSC methods often involve measuring both mass loss and heat flow simultaneously during a continuous temperature ramp, our study employs a two-step approach to achieve more precise control over water vapor content. In the first step, we use TGA to selectively remove and measure the mass of adsorbed water at specific temperatures, ensuring that each sample achieves a well-defined and stable hydration level before analysis. This is followed by a separate DSC run at the defined hydration state, where we only observe the heat flow related to the sample's specific heat capacity, while the water vapor content remains constant within each measurement. This separate approach allows us to better decouple the effects of temperature and water vapor content on MOF-303's heat capacity, providing more accurate insights into the role of water loading without the interference of continuous desorption. By contrast, a conventional TGA-DSC experiment would directly link mass loss with heat flow changes in a single experiment, potentially confounding the interpretation of heat capacity at different hydration stages.

In contrast to simultaneous TGA-DSC, which continuously monitors both mass change and heat flow in one heating ramp, our approach involves two distinct steps to ensure stable, well-defined water contents in each sample before measuring Cp. First, we use a TGA device to systematically heat samples to specific temperatures, thereby controlling the degree of water desorption. We then seal each sample-now at a precisely known water loading-and transfer it to a conventional DSC for heat-capacity measurements. This two-step methodology avoids continuous water loss during the DSC run and ensures that each measured Cp value corresponds to a fixed, reproducible hydration state, rather than tracking dynamic mass changes in real time. Such separation of TGA and DSC experiments clarifies the role of water content in elevating Cp while mitigating overlapping thermal events that can complicate interpretation in simultaneous TGA-DSC.

7. Results and discussion

To investigate the values of Cp for MOF-303 under different water loading, seven sealed crucibles of MOF-303 containing different water vapor quantities were well prepared and tested using DSC-60 Plus devices. The presented section is devoted to showing the results of each MOF-303 sample with different contained water vapor. Fig. 5 presents the DSC heat signal as a function of time and temperature for MOF-303 at varying water vapor loadings. The x-axis represents time in seconds, while the left y-axis corresponds to the DSC heat signal (in mW), and the right y-axis shows the temperature in °C. The black line represents the heating rate, showing a gradual temperature rise from 5 °C to around 95 °C with a rate of 10 °C/min. The colored curves depict DSC signals for different levels of water vapor saturation: from 0 % (zero uptake) to 100 % (highest uptake). The fully saturated MOF (q = 100 %) exhibits the largest endothermic peak, indicating significant water desorption as temperature increases. Lower uptake levels, represented by the other curves (83.4 %, 74.4 %, 34.25 %, 12.02 %, 0.48 %, and 0 %), exhibit progressively smaller endothermic dips, correlating with decreasing water content. The baseline (red line) provides data correction for the standard Al_2O_3 and sample crucibles, while the dashed blue line shows the standard for Al_2O_3, which serves as a stable reference material. The plot demonstrates that the heat of adsorption (endothermic peaks) increases with higher water vapor loading, indicating stronger water interactions in the MOF-303 structure as humidity increases.

7.1. Effect of temperature change at different water vapor loading

Fig. 6 illustrates the Cp of MOF-303 across a range of temperatures (from 10 °C to 95 °C) at different water vapor loadings, denoted by the parameter *q*, which ranges from 0 (completely dry) to 1 (fully saturated). The x-axis represents temperature, while the y-axis shows (Cp) in J/g.°C, spanning from 0.9 to 2.5 J/g.°C. The dry MOF-303 sample (q = 0, black curve) has the lowest specific heat capacity, starting at approximately 0.9175 J/g.°C at 10 °C and increasing gradually to around 1.1695 J/g.°C at 95 °C. This indicates that in the absence of water, the material adsorbed relatively little heat with increasing temperature. On the contrary, as the water vapor loading increased, the specific heat capacity values became significantly larger, reflecting the enhanced heat

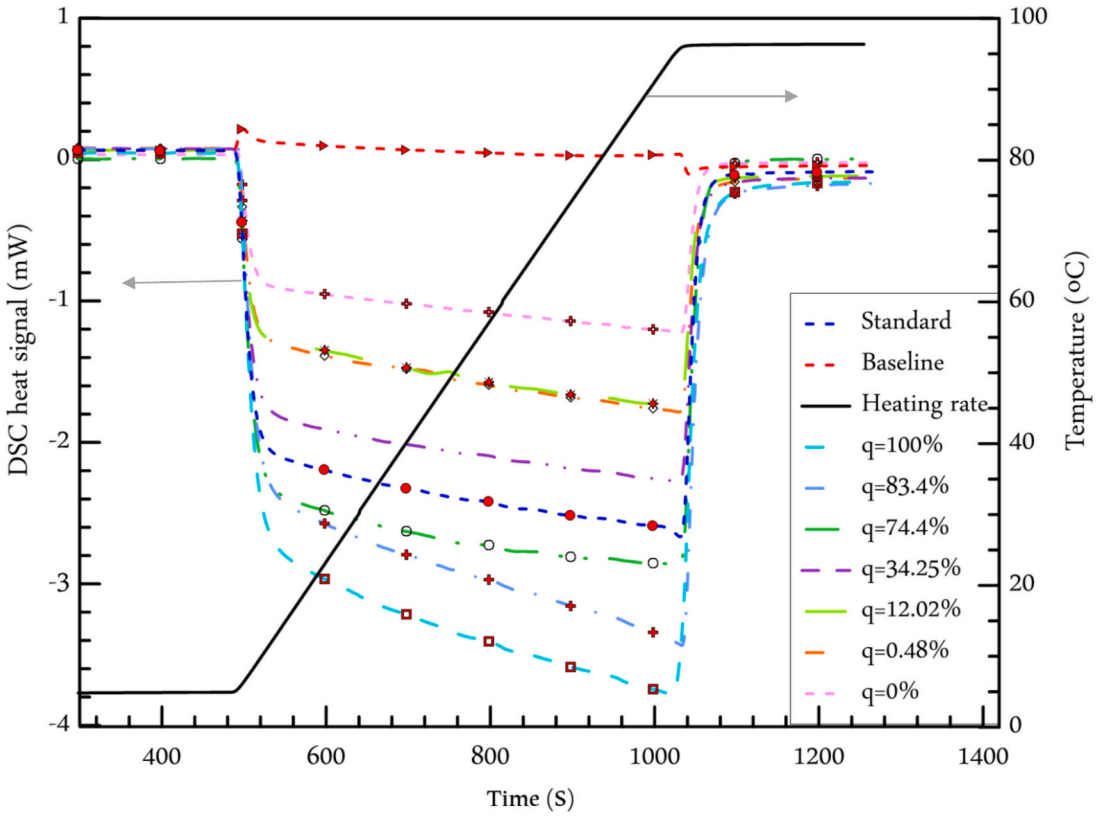


Fig. 5. Heat signals sent by DSC for each sample.

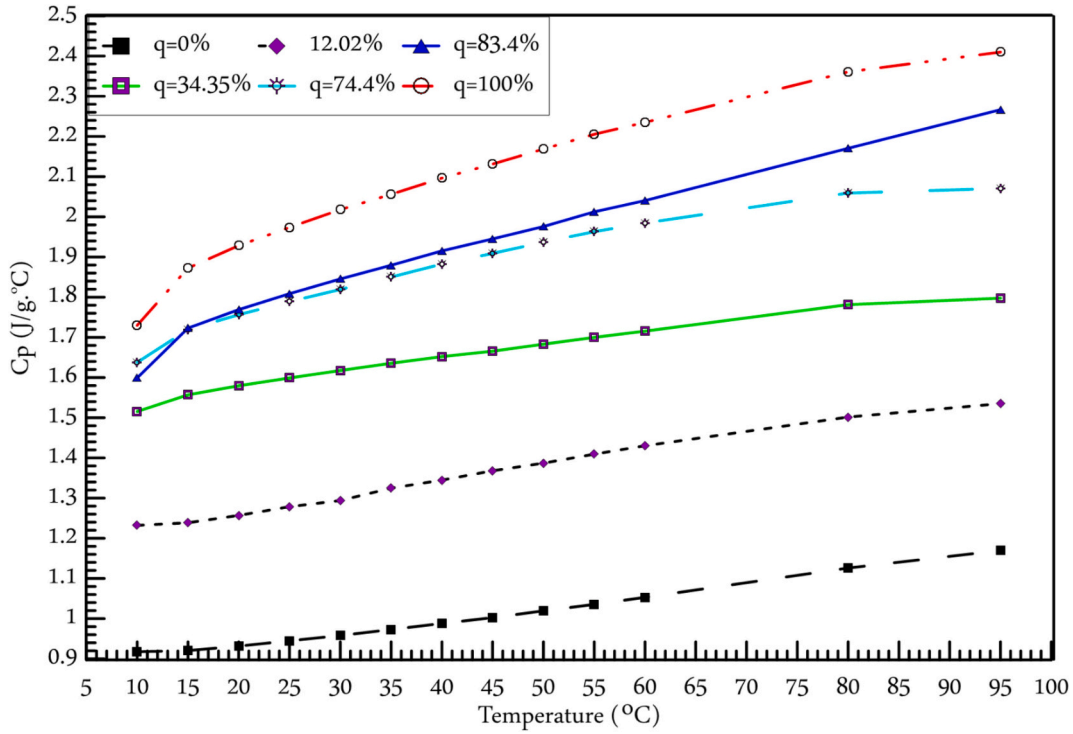


Fig. 6. Specific heat values of different MOF-303 samples at different temperatures.

adsorption due to the presence of water within the MOF structure. For instance, at a low water vapor loading (q = 0.0048, red curve), the Cp began at 1.25 J/g.°C at 10 °C and increased steadily to 1.5305 J/g.°C at 95 °C, indicating that even a small amount of water significantly enhances the heat capacity of the material. Similarly, for intermediate water vapor loadings (q = 0.48), the specific heat capacity starts at around 1.5 J/g.°C and rises to approximately 1.79 J/g.°C by 95 °C, showing that moderate water content also leads to a substantial increase

in heat adsorption capacity. At higher water vapor loadings (q = 0.74 and q = 0.839, cyan and blue curves), the specific heat capacity starts at about 1.6–1.7 J/g.°C and increases to around 2.1–2.2 J/g.°C by 95 °C. These curves show that as the water content approaches full saturation, the Cp values become more pronounced, reflecting stronger interactions between the water molecules and the MOF-303, leading to greater thermal energy adsorption. The fully saturated sample (q = 1, red curve) exhibits the highest specific heat capacity, starting at 1.72 J/g.°C at 10 °C and rising to 2.4 J/g.°C at 95 °C. This steep increase in Cp indicates that when the MOF-303 is completely loaded with water vapor, it adsorbs significantly more heat as the temperature rises, likely due to the desorption and interaction of water molecules within the MOF-303 framework. Fig. 6 demonstrates a clear trend: the specific heat capacity of MOF-303 increases with both temperature and water vapor loading. Dry samples exhibit lower heat capacities, while samples with higher water content show much larger Cp values, indicating that water plays a critical role in enhancing the material's thermal properties. This behavior is particularly relevant for applications where heat management and thermal adsorption are important, as MOF-303's ability to store and release heat depends strongly on its water vapor content.

The specific heat capacity (Cp) of MOF-303 shows a pronounced increase with rising temperature at specific water vapor, a trend consistent with that observed by Kloutse et al. [10] for other frameworks such as MOF-5, Cu-BTC, Fe-BTC, MOF-177, and MIL-53(Al). As temperature elevates, more vibrational, rotational, and translational modes within the MOF's lattice become thermally accessible, thereby allowing the material to store greater amounts of energy. In essence, the progressive activation of these modes contributes to a monotonic rise in Cp, reflecting the thermodynamic behavior common to many MOFs.

7.2. Effect of water vapor loading

Fig. 7 shows the specific heat capacity (Cp) of MOF-303 as a function of water vapor uptake (in kg/kg) at various temperatures, ranging from 10 °C to 95 °C. The x-axis represents the uptake of water in the MOF structure, while the y-axis shows Cp in J/g.°C, ranging from 0.9 to 2.5 J/g.°C. Each color symbol represents a different temperature, with red circles for 10 °C, black circles for 25 °C, purple diamonds for 15 °C, yellow squares for 95 °C, and so on. Fig. 7 indicates that Cp increases with both water uptake and temperature. At the lowest uptake levels (around 0 to 0.1 kg/kg), all temperatures show relatively low specific heat capacities, with values around 0.9175 to 1.1695 J/g.°C. As water uptake increases, the heat capacity also rises sharply, particularly for higher temperatures. For instance, at the highest uptake level (close to 100 kg/kg), the Cp at 95 °C reaches about 2.4 J/g.°C, while at lower temperatures (e.g., 10 °C), Cp increases more moderately, reaching only about 1.6 J/g.°C.

The inset graphs zoom in on specific uptake ranges, providing more detailed views of how the heat capacity changes for smaller increments of water uptake. In the first inset (uptake range 0 to 0.5 kg/kg), it is clear that even a small amount of water significantly increases the heat capacity, particularly at higher temperatures. In the second inset (uptake range 10 to 40 kg/kg), the pattern of rising Cp with increasing temperature and water content is further reinforced, with noticeable separation between curves representing different temperatures. The graph demonstrates that the specific heat capacity of MOF-303 increases as both water uptake and temperature rise, with fully saturated samples at high temperatures showing the highest Cp values. This underscores the importance of water content and temperature in determining the thermal properties of MOF-303, as more water leads to greater heat adsorption. In addition to temperature effects, the water vapor loading in MOF-303 markedly boosts its specific heat capacity. This phenomenon stems from two factors: (1) water itself has a relatively large Cp, thus directly augmenting the material's thermal mass; and (2) the adsorbed molecules introduce new intermolecular interactions, most notably hydrogen bonding, which further expand the range of energy-adsorbing modes available in the composite system. Consequently, fully saturated samples exhibit substantially higher Cp than their dry counterparts, underscoring the critical role water uptake plays in enhancing the thermal properties of MOF-303.

7.3. Repeatability and reproducibility of the experimental results

Fig. 8 provides a detailed overview of repeatability and

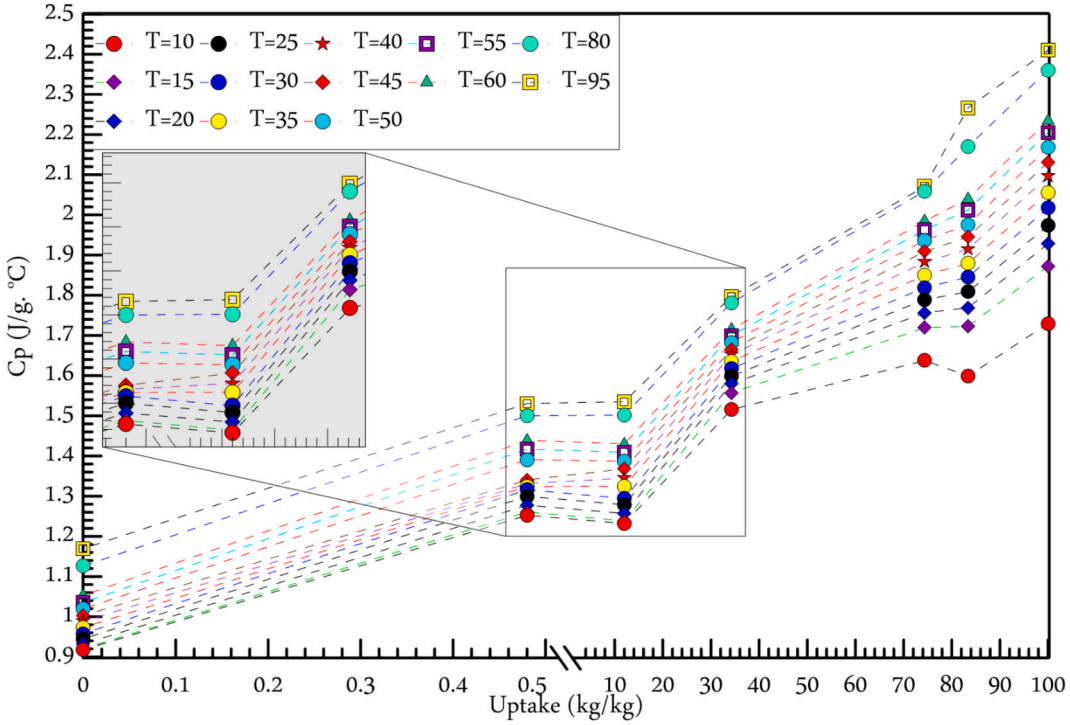


Fig. 7. MOF-303 specific heat values at different water vapor loading.

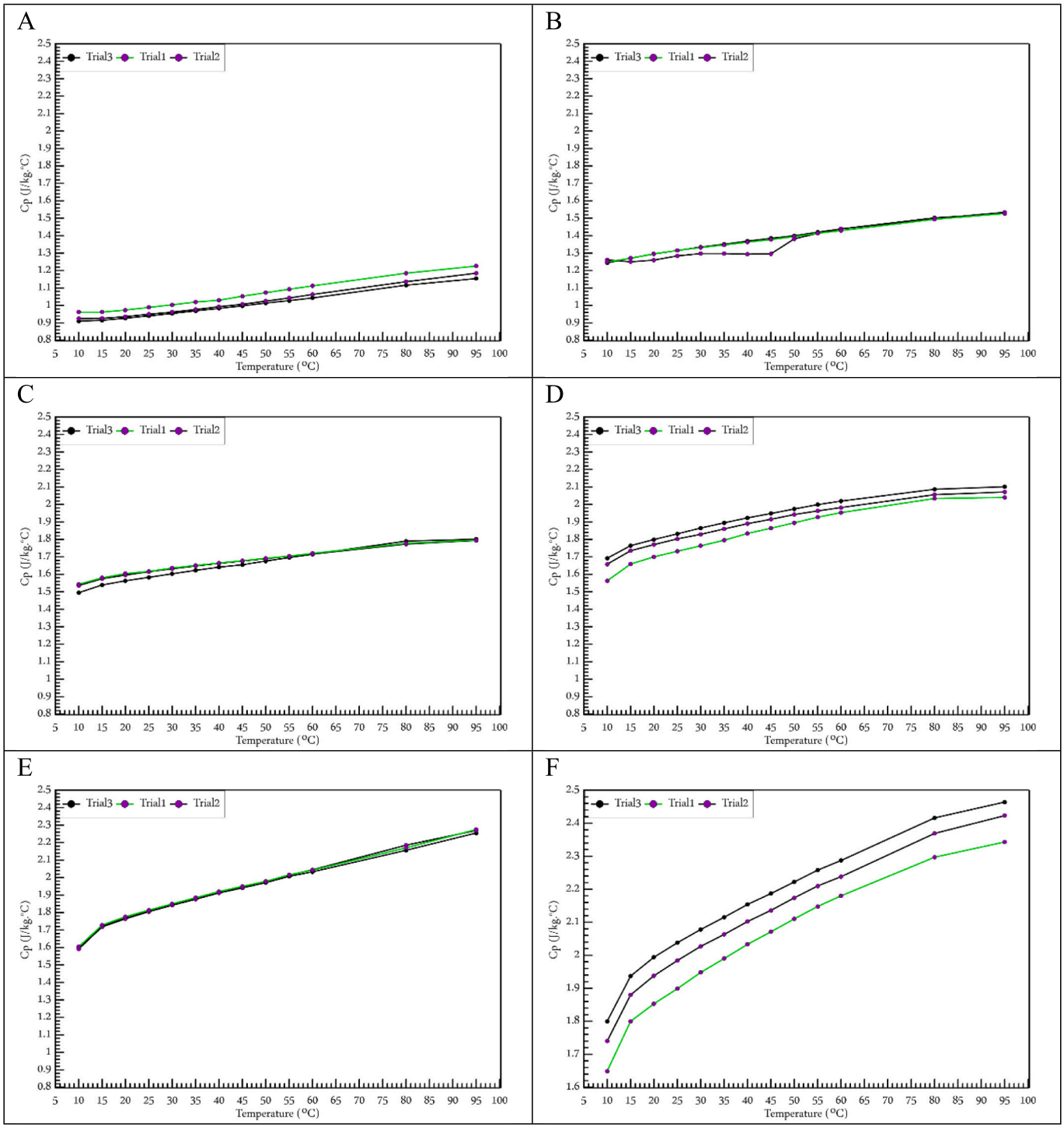


Fig. 8. Repeatability and reproducibility of MOF-303's specific heat measurements across different water vapor loadings: **(A)** q = 0 %, **(B)** q = 12.02 %, **(C)** q = 34.25 %, **(D)** q = 74.44 %, **(E)** q = 83.42 %, and **(F)** q = 100 %.

reproducibility in MOF-303's measured Cp at temperatures ranging from 10 °C to 95 °C and at six distinct water vapor loadings (q = 0 %, 12.02 %, 34.25 %, 74.44 %, 83.42 %, and 100 %). This figure underscores the measurement consistency attained through multiple DSC runs, with most standard deviations remaining below 5 % of the corresponding mean values strongly indicative of robust reproducibility. Even for the fully saturated sample at 80 °C (q = 100 %), the average Cp of around 1.126 J/g.°C is associated with a tight standard deviation of approximately 0.014, confirming both the reliability of the DSC procedure and the pronounced effect of water vapor loading on MOF-303's heat storage properties.

Additionally, each sub-figure presents multiple DSC measurements (three per temperature). Notably, at 10 °C, the dry sample (q = 0 %) exhibits an average Cp of 0.9175 ± 0.0106 J/g.°C, whereas the fully saturated sample (q = 100 %) displays a significantly higher average Cp of 1.7297 ± 0.0760 J/g.°C. As temperature increases, all samples experience elevated Cp values, although the magnitude of the increase depends on the water vapor loading. For instance, at 95 °C, the dry sample's Cp reaches 1.1695 ± 0.0219 J/g.°C, while the fully saturated sample approaches 2.41 ± 0.06 J/g.°C, signifying a substantial

enhancement in heat adsorption capacity due to adsorbed water. In addition to highlighting the strong dependence of Cp on both temperature and water vapor content, these data provide valuable evidence for the robustness of the DSC procedure in capturing subtle yet significant variations in heat capacity.

7.4. Specific heat correlations

7.4.1. Specific heat correlations as a function of temperature

Concluding the results from measuring specific heat values could be obtained in experimental correlations as a function of temperature for each water vapor loading and thus double interpolation could be implemented to obtain the specific heat values at not given water vapor loading and temperature values or using machine learning. Fig. 9 highlights the fitting output data for each data set and the corresponding equations and statics. A polynomial function of order 3 could be applied to fit the data at constant water vapor loading. Table 3 finalizes the obtained correlations. These equations could be applied to the estimated Cp of MOF-303 in a temperature range between 5 and 95 °C. Other information should be stated to the leader such as MOF-303 is solvothermal synthesized at a small scale. Following the synthesis, MOF-303 characterization has been published [16] and put in the supplementary data file as well.

7.5. Specific heat correlation as a function of temperature and water vapor loading

The provided 3D Fig. 10 illustrates the variation Cp for MOF-303 as a function of temperature and adsorption uptake q at different levels of saturation. The red data points correspond to experimentally measured values of Cp , while the cyan surface represents the fitting curve generated using a power 2D model in Origin software. The uptake values q considered in the model range from fully saturated to nearly dry, with q = 1 indicating full saturation, followed by q = 0.834, 0.744, 0.3425, 0.1202, 0.0048, and q = 0, corresponding to progressively lower adsorbate loadings. Temperature varies from 10 °C to 95 °C. The graph reveals that as the temperature increases, Cp tends to increase non-linearly, with more pronounced curvature at higher uptake levels, demonstrating the significant dependence of Cp on both temperature and the degree of adsorption. The fitted surface provides a smooth transition between the data points, indicating the accuracy of the power 2D model in capturing the relationship between the variables with the

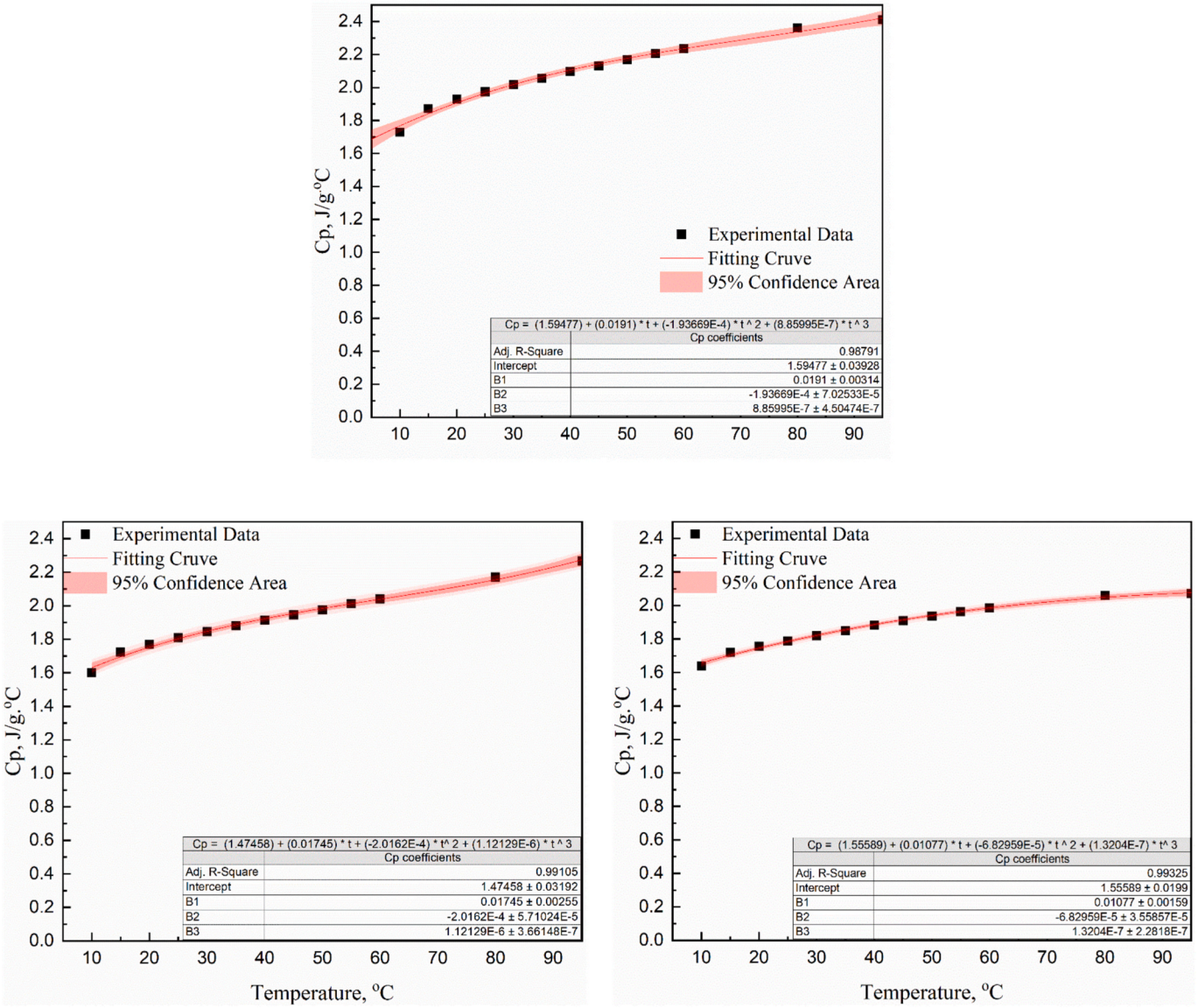


Fig. 9. Fitting Curves of MOF-303 specific heat.

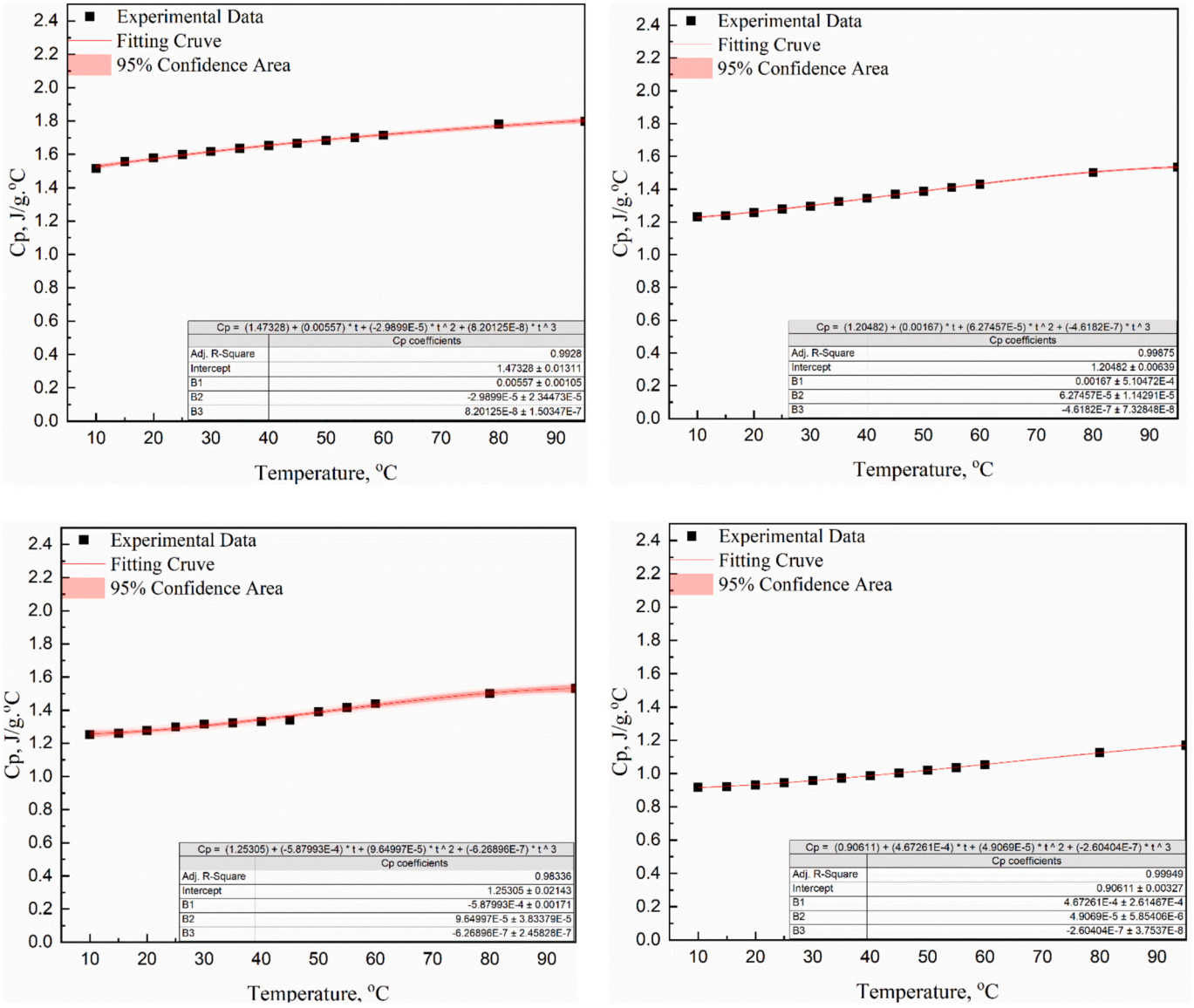


Fig. 9. (*continued*).

Table 3
MOF-303 Specific heat correlations as a function of temperature.

$c_{p_{q=100}} = 1.59477 + 0.0191*T^1 + (-1.93669E-4)*T^2 + (8.85995E-7)*T^3$	Eq. (3)
$c_{p_{q=83.4}} = 1.47458 + 0.01745*T^1 + (-2.0162E-4)*T^2 + (1.12129E-7)*T^3$	Eq. (4)
$c_{p_{q=74.4}} = 1.55589 + 0.01077*T^1 + (-6.82959E-5)*T^2 + (1.3204E-7)*T^3$	Eq. (5)
$c_{p_{q=34.25}} = 1.47328 + 0.00557*T^1 + (-2.9899E-5)*T^2 + (8.20125E-8)*T^3$	Eq. (6)
$c_{p_{q=12.02}} = 1.20482 + 0.00167*T^1 + (6.27457E-5)*T^2 + (-4.6182E-7)*T^3$	Eq. (7)
$c_{p_{q=0.48}} = 1.25305 + (-5.87993E-4)*T^1 + (9.64997E-5)*T^2 + (-6.26896E-7)*T^3$	Eq. (8)
$c_{p_{q=0}} = 0.90611 + (4.67261E-4)*T^1 + (4.9069E-5)*T^2 + (-2.60404E-7)*T^3$	Eq. (9)

root mean square (R = 93.42 %). The trend also suggests that Cp rises more steeply at higher adsorption levels, reflecting the thermodynamic response of MOF-303 under different thermal and adsorption conditions. Table 4 highlights the Equations derived from the fitting model and their conditions. In an adsorption cooling cycle-an inherently thermal and mass-transfer system-the parameters most influential in performance include specific heat capacity (Cp), thermal conductivity, heat transfer coefficient, and diffusion coefficient [22–24]. Specific heat, in particular, exerts opposing influences on adsorption and desorption processes. During adsorption, a higher Cp means the bed temperature rises more slowly in response to the heat of adsorption, favoring enhanced uptake (a beneficial effect). Conversely, in desorption, a higher Cp requires more energy to elevate the bed temperature sufficiently to drive off the adsorbate (an adverse effect). Accordingly, characterizing Cp as a function of temperature and water vapor loading is critical for optimizing cycle design—helping determine appropriate heating/cooling water temperatures, proper adsorbent mass, and overall operational strategies.

7.6. Thermal stability of MOF-303 specific heat values

The thermal stability of MOF-303 specific heat values is investigated using the cyclic Saphire method with the same procedure for both heating and cooling. The heat rate applied is 10 °C/min to raise the

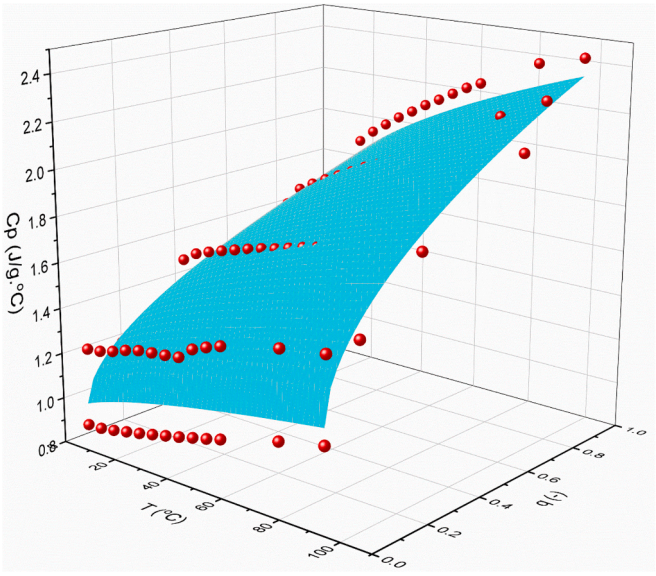


Fig. 10. MOF-303's Cp fitting curves in terms of temperature and uptake.

Table 4
MOF-303 Specific heat correlations as a function of temperature and water vapor loading.

New correlation: $C_p = A + BT^C + Dq^E + FT^C q^E$	
Conditions: $10^\circ C \leq T \leq 95^\circ C$	
$0 \leq q \leq 1, q$ *is maximum when 30%uptake*	
MOF-303 is a solvothermal synthesized	
A	0.91115 ± 0.11356
B	0.03069 ± 0.04405
C	0.52139 ± 0.25086
D	0.59655 ± 0.17041
E	0.51417 ± 0.0439
F	0.046 ± 0.06606
Adj. R-Square	0.93424

temperature of the MOF sample from 5 °C to 95 °C and the temperature is stabilized for 8 mins before being cooled with the same rate of 10 °C/min down to 5 °C with stabilizing for 4 mins. The test covers approximately 100 min to ensure that the sample uptake and intake consume the same heat signals and thus the same specific heat values. Fig. 11 indicates the heat signal sent to the MOF-303 sample at a given time instant and the corresponding temperature value. MOF-303 sample adsorbs heat with a rate up to 3 mW nearly to 95 °C and extracts the same heat rate cooling down nearly to 5 °C, therefore, the same specific heat values are obtained in the case of heating and cooling, logically, sensible heating is a reversible process. Additionally, excellent thermal stability for MOF-303 at a specific water vapor loading is observed not only for the peaks and minimum values for heat signals but also for the behavior at each process for heating and cooling.

It is worth mentioning that adding graphene and carbon nanotubes (CNTs) could enhance the thermal conductivity of MOF-303 by providing additional percolation pathways and bridging interface gaps where heat transfer would otherwise be hindered. High-conductivity carbon additives, such as graphene and carbon nanotubes, can effectively form conductive channels throughout the MOF framework, mitigating the intrinsic low thermal diffusivity of MOF-303. Recent work on MOF-303/graphene adsorbents [25] illustrates how even modest loadings of graphene dramatically increase thermal diffusivity, thereby shortening the heating–cooling cycle time in adsorption applications. Likewise, incorporating CNTs into MOFs for thermoelectric devices [26] has been shown to reduce interfacial thermal resistance and improve overall mechanical stability. Taken together, these studies indicate that MOF-303-carbon composites offer a promising route to more efficient heat management and faster adsorb–desorb cycles.

7.7. Specific heat comparison using other published data

Fig. 12 compares the temperature-dependent specific heat capacity (Cp) of five different MOFs in the dry state, MOF-303, MIL-53(Al), MOF-5, Fe-BTC, and MOF-177, over a range of roughly 10 °C to 45 °C. MOF-303 consistently exhibits the highest Cp, beginning near 0.91 J/g.°C at 10 °C and rising to about 1.00 J/g.°C at 45 °C, followed closely by MIL-53(Al), which ranges from approximately 0.80 J/g.°C at 10 °C to 0.91 J/

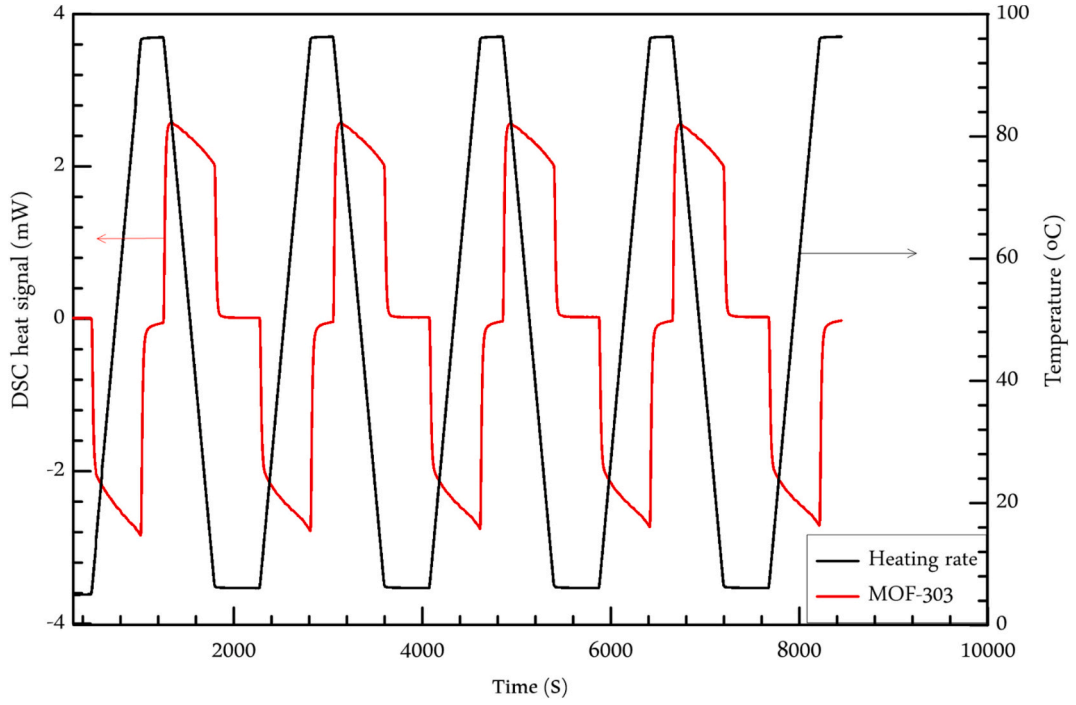


Fig. 11. MOF-303's Cp thermal stability test.

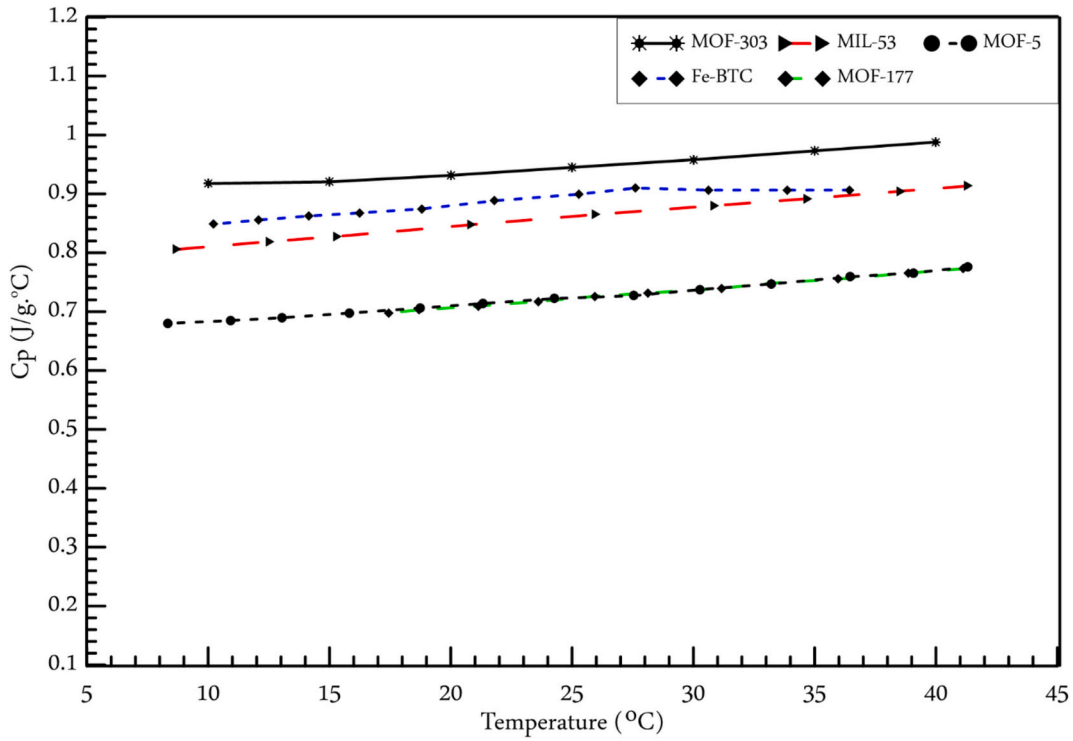


Fig. 12. MOF-303's Cp comparison with other different materials [14].

g.°C at 41 °C. Fe-BTC occupies a mid-range (0.84 to 0.90 J/g.°C), while both MOF-5 and MOF-177 display comparatively lower values (0.68 to 0.78 J/g.°C). These differences largely reflect each MOF's metal center (Al vs. Zn, Fe), linker chemistry (triazolate, terephthalate, etc.), and structural features (porosity, density, flexibility).

Higher Cp profiles in MOF-303 and MIL-53(Al) stem from their Al-based nodes coupled with bridging hydroxyl groups, which introduce additional vibrational modes. The linkers-1,2,4-triazolate (MOF-303) and terephthalate (MIL-53(Al))-impart greater structural flexibility than more rigid linkers found in MOF-5 and MOF-177, thus enhancing heat adsorption. Nevertheless, direct comparisons between Al- and Zn-based frameworks can be oversimplified unless one also considers pore geometry, guest molecules, and the mass contributions of the linkers. Although pure aluminum has a higher specific heat capacity than zinc, the complex interplay of organic linkers, metal clusters, and porosity ultimately determines the overall Cp. As such, the elevated Cp observed for MOF-303 and MIL-53(Al) highlights their potential for adsorption-based cooling systems or other thermal applications, whereas MOF-5, MOF-177, and Fe-BTC may be optimized for different performance metrics (e.g., gas storage capacity).

8. Conclusions

This study presents the first detailed measurements of the specific heat capacity (Cp) of MOF-303, for a wide range of temperature and water vapor content. The results show that

1- The Cp of MOF-303 increases substantially with water vapor loading, demonstrating its potential as an efficient material for adsorption cooling systems.
2- Dry MOF-303 samples exhibited a Cp ranging from 0.9175 J/g.^{o}C at 10 °C to 1.1695 J/g.^{o}C at 95 °C
3- Fully saturated samples showed higher values, with Cp reaching 1.72 J/g.^{o}C at 10 °C and 2.41 J/g.^{o}C at 95 °C.
4- For the first time, a new correlation for Cp as a function of temperature, and water vapor uptake has been derived; providing a valuable tool for predicting MOF-303's thermal behavior under varying conditions and based on the specifications of sythnizing MOF-303.
5- Thermal cyclic stability tests confirmed that MOF-303 maintains consistent Cp values under repeated heating and cooling cycles, reinforcing its long-term reliability in practical applications.
6- These findings not only offer new insights into the thermal properties of MOF-303 but also lay the groundwork for optimizing adsorption-based cooling systems.
7- A promising direction for future research is to synthesize and characterize the specific heat capacity (Cp) of MOF-303/CNT and MOF-303/graphene composites. While pristine MOF-303 exhibits excellent adsorption capabilities, its inherently low thermal conductivity can prolong heating and cooling cycles. Incorporating high-conductivity carbon additives-whether CNTs or graphene-can help alleviate this limitation. Recent studies on MOF/CNT thermoelectric hybrids [26] show that coupling MOFs with CNTs not only mitigates thermal resistance via heterogeneous interfaces but also enhances mechanical stability and charge transport. Likewise, work on MOF-303/graphene composites for wire-finned heat exchangers [25] demonstrates a substantial rise in thermal diffusivity, enabling more efficient adsorb-desorb cycles. Together, these findings underscore the promise of MOF-303/CNT or graphene-based for applications requiring both robust adsorption and improved heat transfer.

CRediT authorship contribution statement

Ahmed E. Abu El-maaty: Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ourida Saoudi:** Writing – original draft, Validation, Software, Data curation. **Ahmed S. Abdelrazik:** Writing – original draft, Supervision, Software, Resources, Methodology, Data curation. **Rached Ben-Mansour:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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